Numerical Solution of ordinary Differential Equation

Taylor's Series method.

Consider the first order differential Equation

$$\frac{dy}{dx} = f(x,y) \quad \text{with the initial condition } y(x_0)=y_0 \qquad \rightarrow (1)$$

If $y(x)$ is the exact solution of (1), then the Taylor's Series for $y(x)$ around $x=x_0$ is given by

$$y(x) = y_0 + (x-x_0)y_0' + \frac{(x-x_0)^2}{2!}y_0'' + \frac{(x-x_0)^3}{3!}y_0''' + \cdots$$

(1) Find by Taylor's Series method the value of y at $x=0.2$ correct to four decimal places if $y(x)$ satisfies $\frac{dy}{dx} = x-y^2$ and $y(0)=1$.

↳ $x=0, y=1$ Here $x_0=0, y_0=1$

Soln:- we have

$$\frac{dy}{dx} = x-y^2 \qquad \Rightarrow y_0' = 0-1^2 = -1$$

$$y' = x-y^2$$

Diff above eqn, we get

$$y'' = 1-2yy' \qquad \Rightarrow y_0'' = 1-2y_0y_0' = 1-2(1)(-1) = 3$$

$$y''' = -2yy''-2(y')^2 \qquad \Rightarrow y_0''' = -2(y_0)(y_0'')-2(y_0')^2 = -2(1)(3)-2(-1)^2 = -8$$

$$y^{iv} = -2yy'''-2y'y''-4y'y'' \qquad \Rightarrow y_0^{iv} = -2(1)(-8)-2(-1)(3)-4(-1)(3) = 34$$

$$y^v = -2\left[yy^{iv} + y'y''' + 2y'y''' + 2(y'')^2\right] \Rightarrow y_0^v = -186$$

$\therefore$ By Taylor's series we have

$$y(x) = y_0 + (x-x_0)y_0' + \frac{(x-x_0)^2}{2!}y_0'' + \frac{(x-x_0)^3}{3!}y_0''' + \frac{(x-x_0)^4}{4!}y_0^{iv} + \cdots$$

$$= 1 + (x-0)(-1) + \frac{(x-0)^2}{2}(3) + \frac{(x-0)^3}{6}(-8) + \frac{(x-0)^4}{24}(34) + \frac{(x-0)^5}{120}(-186) + \cdots$$

$$= 1 - x + \frac{3}{2}x^2 - \frac{4}{3}x^3 + \frac{17}{12}x^4 - \frac{31}{20}x^5 + \cdots$$

Put $x = 0.2$

$$y(0.2) = 1 - 0.2 + \frac{3}{2}(0.2)^2 - \frac{4}{3}(0.2)^3 + \frac{17}{12}(0.2)^4 - \frac{31}{20}(0.2)^5 + \cdots$$

$$= 0.8511$$

(2) Employ the Taylor's method to obtain approximate value of y at $x = 0.2$ for the differential equation $\frac{dy}{dx} = 2y + 3e^x$ with $y(0) = 0$.

Soln: Given $\frac{dy}{dx} = 2y + 3e^x$ $\quad y(0) = 0 \Rightarrow x_0 = 0, y_0 = 0$

$$y' = 2y + 3e^x \Rightarrow y_0' = 2(0) + 3e^0 = 3$$

$$y'' = 2y' + 3e^x \Rightarrow y_0'' = 2(3) + 3e^0 = 9$$

$$y''' = 2y'' + 3e^x \Rightarrow y_0''' = 2(9) + 3e^0 = 21$$

$$y^{iv} = 2y''' + 3e^x \Rightarrow y_0^{iv} = 2(21) + 3e^0 = 45$$

∴ By Taylor's series method, we have

$$y(x) = y_0 + (x-x_0)y_0' + \frac{(x-x_0)^2}{2!}y_0'' + \frac{(x-x_0)^3}{3!}y_0''' + \frac{(x-x_0)^4}{4!}y_0^{iv} + \cdots$$

$$y(x) = 0 + (x-0)(3) + \frac{(x-0)^2}{2}(9) + \frac{(x-0)^3}{6}(21) + \frac{(x-0)^4}{24}(45) + \cdots$$

$$= 3x + \frac{9x^2}{2} + \frac{21}{6}x^3 + \frac{45}{24}x^4 + \cdots$$

Put $x = 0.2$

$$y(0.2) = 3(0.2) + \frac{9}{2}(0.2)^2 + \frac{21}{6}(0.2)^3 + \frac{45}{24}(0.2)^4 + \cdots$$

$$= 0.8110$$

(3) Find the solution of the initial value problem $\frac{dy}{dx} = y^2 + x$, $y(0) = 1$ at the points $x = 0.1, 0.2$ and 0.3.

($x_0 = 0$)

Soln: we have $\frac{dy}{dx} = y^2 + x$, $x_0 = 0$, $y_0 = 1$

$y' = y^2 + x$ $\Rightarrow y_0' = 1^2 + 0 = 1$

$y'' = 2yy' + 1$ $\Rightarrow y_0'' = 2(1)(1) + 1 = 3$

$y''' = 2yy'' + 2(y')(y')$ $\Rightarrow y_0''' = 2(1)(3) + 2(1)^2$
$= 8$

$y^{iv} = 2[yy''' + 3y'y''] + 2y'y'']$

$= 2[yy''' + 3y'y'']$ $\Rightarrow y_0^{iv} = 2[(1)(8) + 3(1)(3)]$
$= 34$

By Taylor's series method we have

$$y(x) = y_0 + (x-x_0)y_0' + \frac{(x-x_0)^2}{2!}y_0'' + \frac{(x-x_0)^3}{3!}y_0''' + \frac{(x-x_0)^4}{4!}y_0^{iv} + \cdots$$

$$y(x) = 1 + (x-0)(1) + \frac{(x-0)^2}{2}(3) + \frac{(x-0)^3}{6}(8) + \frac{(x-0)^4}{24}(34) + \cdots$$

$$= 1 + x + \frac{3x^2}{2} + \frac{4}{3}x^3 + \frac{17}{12}x^4 + \cdots \rightarrow (1)$$

Put $x = 0.1$ in (1), we get

$$y(0.1) = 1 + 0.1 + \frac{3}{2}(0.1)^2 + \frac{4}{3}(0.1)^3 + \frac{17}{12}(0.1)^4 + \cdots$$

$$= 1.1164$$

Put $x = 0.2$ in (1), we get

$$y(0.2) = 1 + 0.2 + \frac{3}{2}(0.2)^2 + \frac{4}{3}(0.2)^3 + \frac{17}{12}(0.2)^4 + \cdots$$

$$y(0.2) = 1.2729$$

Put $x = 0.3$ in (1), we get

$$y(0.3) = 1 + 0.3 + \frac{3}{2}(0.3)^2 + \frac{4}{3}(0.3)^3 + \frac{17}{12}(0.3)^4 + \cdots$$

$$y(0.3) = 1.4824$$

(4) Using Taylor's Series Solve the initial value problem

$\frac{dy}{dx} = xy - 1$ with $y(1) = 2$ at the points $x = 1.02, 1.04$

Soln: Here $\frac{dy}{dx} = xy - 1$, $y_0 = 2$, $x_0 = 1$

$$y' = xy - 1 \Rightarrow y_0' = (1)(2) - 1 = 1$$

$$y'' = xy' + y \Rightarrow y_0'' = (1)(1) + 2 = 3$$

$$y''' = xy'' + y' + y' \Rightarrow y_0''' = (1)(3) + 1 + 1 = 5$$

$$y''' = xy'' + 2y'$$

$$y^{iv} = xy''' + y'' + 2y''$$

$$= xy''' + 3y'' \Rightarrow y_0^{iv} = (1)(5) + 3(3) = 14$$

By the Taylor's Series method we have

$$y(x) = y_0 + (x-x_0)y_0' + \frac{(x-x_0)^2}{2!}y_0'' + \frac{(x-x_0)^3}{3!}y_0''' + \frac{(x-x_0)^4}{4!}y_0^{iv} + \cdots$$

$$= 2 + (x-1)(1) + \frac{(x-1)^2}{2}(3) + \frac{(x-1)^3}{6}(5) + \frac{(x-1)^4}{24}(14) + \cdots$$

$\rightarrow$ (1)

Put $x = 1.02$ in (1), we get

$y(1.02) = 2.0206$

Put $x = 1.04$ in (1), we get

$y(1.04) = 2.0424.$

(5) Using Taylor's Series find the solution of $x\frac{dy}{dx} = x - y$, $y(2) = 2$ at $x = 2.1$

Soln: Here $x_0 = 2$ and $y_0 = 2$

The D.Eqn is $y' = \frac{x-y}{x}$

$$y' = 1 - \frac{y}{x} \Rightarrow y'(2) = 1 - \frac{2}{2} = 0$$

$$y'' = -\frac{y'}{x} + \frac{y}{x^2} \Rightarrow y_0'' = -\frac{0}{2} + \frac{2}{2^2} = \frac{1}{2}$$

$$y''' = -\frac{y''}{x} + \frac{y'}{x^2} + \frac{y'}{x^2} - \frac{2y}{x^3}$$

$$= -\frac{y''}{x} + \frac{2y'}{x^2} - \frac{2y}{x^3} \Rightarrow y_0''' = -\frac{1}{2(2)} + 0 - \frac{4}{8}$$

$$= -\frac{3}{4}$$

$$y^{iv} = -\frac{y'''}{x} + \frac{y''}{x^2} + \frac{2y''}{x^2} - \frac{4y'}{x^3} - \frac{2y'}{x^3} + \frac{6y}{x^4}$$

$$= -\frac{y'''}{x} + \frac{3y''}{x^2} - \frac{6y'}{x^3} + \frac{6y}{x^4} \Rightarrow y_0^{iv} = \frac{3}{8} + \frac{3}{8} - 0 + \frac{12}{16} = \frac{3}{2}$$

$\therefore$ By Taylor's Series we have

$$y(x) = 2 + (x-2)(0) + \frac{(x-2)^2}{2}\left(\frac{1}{2}\right) + \frac{(x-2)^3}{8}\left(\frac{3}{4}\right) + \frac{(x-2)^4}{24}\left(\frac{3}{2}\right)$$

$$= 2 + \frac{(x-2)^2}{4} - \frac{(x-2)^3}{8} + \frac{(x-2)^4}{16} + \cdots$$

Put $x = 2.1$, we get

$$y(2.1) = 2 + \frac{(0.1)^2}{4} - \frac{(0.1)^3}{8} + \frac{(0.1)^4}{16} - \cdots$$

$$= 2.0023.$$

Improved
OR
Modified Euler's Method. (Predictor-Corrector Method)

Consider the initial value problem

$$\frac{dy}{dx} = f(x,y) \quad , \quad y(x_0) = y_0$$

Euler's method Formula:- (Predictor method)

$$y_{n+1}^{(E)} = y_n + hf(x_n, y_n)$$

Modified Euler's Method Formula:- (Corrector method)

$$y_{n+1} = y_n + \frac{h}{2}\left[f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(E)}) \right]$$

① Using the modified Euler's method, solve the initial value problem $\frac{dy}{dx} = x+y$, $y(0)=1$ at the point $x=0.2$. Take $h=0.2$ and carry out ~~three~~ three modifications.

Soln:- Here $f(x,y) = x+y$

$x_0 = 0$, $y_0 = 1$, $h = 0.2$, we have $x_1 = x_0 + h$

$= 0 + 0.2$

$= 0.2$

we required to find $y_1 = y(x_1) = y(0.2)$

By using Euler's method, we find an approximate value of y at $x_1 = 0.2$

Euler's method $y_1^{(E)} = y_0 + hf(x_0, y_0)$ $\quad \because f(x,y) = x+y$

$= y_0 + h(x_0 + y_0)$ $\quad f(x_0, y_0) = x_0 + y_0$

$= 1 + 0.2(0+1)$

$y_1^{(E)} = 1.2$

Now, the modified Euler's formula gives the first modified value at $x_1 = 0.2$ as

1st Modification

$$y_1^{(1)} = y_0 + \frac{h}{2}\left[f(x_0, y_0) + f(x_1, y_1^{(E)})\right]$$

$$= y_0 + \frac{h}{2}\left[x_0 + y_0 + x_1 + y_1^{(E)}\right]$$

$$= 1 + \frac{0.2}{2}\left[0 + 1 + 0.2 + 1.2\right]$$

$$= 1 + 0.1\,(2.4)$$

$$y_1^{(1)} = 1.24$$

Second modified value of y at $x_1 = 0.2$ is

$$y_1^{(2)} = y_0 + \frac{h}{2}\left[f(x_0, y_0) + f(x_1, y_1^{(1)})\right]$$

$$= y_0 + \frac{h}{2}\left[x_0 + y_0 + x_1 + y_1^{(1)}\right]$$

$$= 1 + \frac{0.2}{2}\left[0 + 1 + 0.2 + 1.24\right]$$

$$= 1 + 0.1\,(2.44)$$

$$y_1^{(2)} = 1.244$$

Third modified value of y at $x_1 = 0.2$ is

$$y_1^{(3)} = y_0 + \frac{h}{2}\left[f(x_0, y_0) + f(x_1, y_1^{(2)})\right]$$

$$= y_0 + \frac{h}{2}\left[x_0 + y_0 + x_1 + y_1^{(2)}\right]$$

$$= 1 + \frac{0.2}{2}\left[0 + 1 + 0.2 + 1.244\right]$$

$$= 1 + 0.1\,(2.444)$$

$$y_1^{(3)} = 1.2444$$

Thus $y(0.2) = 1.2444$ is the required solution.

(2) Using the modified Euler's method solve the initial value problem $\frac{dy}{dx} = \log(x+y)$, $y(1)=2$ at the point $x=1.2$.

Soln:- Here $f(x,y) = \log(x+y)$, $x_0=1$, $y_0=2$, $h=0.2$

$$x_1 = x_0 + h$$
$$= 1+0.2 = 1.2$$

We require to find $y_1 = y(x_1) = y(1.2)$

By Euler's formula, we find an approximate value of y at $x_1=1.2$

$f(x,y) = \log(x+y)$

$f(x_0,y_0) = \log(x_0+y_0)$

$$y_1^{(E)} = y_0 + hf(x_0,y_0)$$
$$= y_0 + h\log(x_0+y_0)$$
$$= 2+0.2\log(1+2)$$
$$= 2+0.2\log 3$$
$$= 2+0.2(1.0986)$$
$$y_1^{(E)} = 2.2197$$

The modified Euler's formula gives the first modified value of y at $x_1=1.2$ as

$$y_1^{(1)} = y_0 + \frac{h}{2}\left[f(x_0,y_0) + f(x_1,y_1^{(E)})\right]$$
$$= y_0 + \frac{h}{2}\left[\log(x_0+y_0) + \log(x_1+y_1^{(E)})\right]$$
$$= 2 + \frac{0.2}{2}\left[\log(1+2) + \log(1.2+2.2197)\right]$$
$$= 2+0.1\left[1.0986 + 1.2295\right]$$
$$y_1^{(1)} = 2.2328$$

Second modified value of y at $x=1.2$

$$y_1^{(2)} = y_0 + \frac{h}{2}\left[f(x_0,y_0) + f(x_1, y_1^{(1)})\right]$$

$$= y_0 + \frac{h}{2}\left[\log(x_0+y_0) + \log(x_1 + y_1^{(1)})\right]$$

$$= 2 + \frac{0.2}{2}\left[\log(1+2) + \log(1.2+2.2328)\right]$$

$$= 2 + 0.1\left[\log(3) + \log(3.4328)\right]$$

$$y_1^{(2)} = 2.2331$$

$\therefore y(1.2) = 2.2331$ is the required solution.

(3) Using the modified Euler's method solve the initial value problem $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0)=1$ at the point $x=0.1$.

Soln:- Here $f(x,y) = \frac{y-x}{y+x}$, $x_0=0$, $y_0=1$, $h=0.1$

$$x_1 = x_0 + h = 0+0.1$$
$$= 0.1$$

By Euler's formula, we have

$$y_1^{(E)} = y_0 + hf(x_0,y_0) \qquad \left[\because f(x,y) = \frac{y-x}{y+x}\right]$$

$$= y_0 + h\left[\frac{y_0-x_0}{y_0+x_0}\right]$$

$$= 1 + 0.1\left[\frac{1-0}{1+0}\right]$$

$$y_1^{(E)} = 1.1$$

The modified Euler's formula gives the first modified value of y.

$$y_1^{(1)} = y_0 + \frac{h}{2}\left[f(x_0,y_0) + f(x_1, y_1^{(E)})\right]$$

$$= y_0 + \frac{h}{2}\left[\frac{y_0 - x_0}{y_0 + x_0} + \frac{y_1^{(E)} - x_1}{y_1^{(E)} + x_1}\right]$$

$$= 1 + \frac{0.1}{2}\left[\frac{1-0}{1+0} + \frac{1.1 - 0.1}{1.1 + 0.1}\right]$$

$$= 1 + 0.05\,[1 + 0.8333]$$

$$y_1^{(1)} = 1.0916$$

$$y_1^{(2)} = y_0 + \frac{h}{2}\left[f(x_0,y_0) + f(x_1, y_1^{(1)})\right]$$

$$= y_0 + \frac{h}{2}\left[\frac{y_0 - x_0}{y_0 + x_0} + \frac{y_1^{(1)} - x_1}{y_1^{(1)} + x_1}\right]$$

$$= 1 + \frac{0.1}{2}\left[\frac{1-0}{1+0} + \frac{1.0916 - 0.1}{1.0916 + 0.1}\right]$$

$$= 1 + 0.05\,[1 + 0.8321]$$

$$y_1^{(2)} = 1.0916$$

Thus the required solution is $y(0.1) = 1.0916$.

(4) Using modified Euler's method, solve the problem

$$\frac{dy}{dx} = \sin x + \cos y, \quad y(2.5) = 0$$ at the points $x = 3$ and $x = 3.5$ in two steps, modifying the solution two times at each step.

Soln:- $f(x,y) = \sin x + \cos y$, $x_0 = 2.5$, $y_0 = 0$

For obtaining the solutions at $x=3$ and $x=3.5$, we take $h=0.5$ so that $x_1=3$ and $x_2=3.5$

We are required to find $y_1=y(x_1)=y(3)$ and $y_2=y(x_2)=y(3.5)$

$$y_1^{(E)} = y_0 + hf(x_0,y_0)$$

$$= y_0 + h(\sin x_0 + \cos y_0)$$

$$= 0 + 0.5\,[\sin 2.5 + \cos 0]$$

$$= 0 + 0.5\,[0.5984 + 1]$$

$$y_1^{(E)} = 0.7992$$

By the Modified Euler's method, the first modified value of y at $x_1=3$ is

$$y_1^{(1)} = y_0 + \frac{h}{2}\left[f(x_0,y_0) + f(x_1, y_1^{(E)})\right]$$

$$= 0 + \frac{0.5}{2}\left[\sin(2.5) + \cos 0 + \sin 3 + \cos(0.7993)\right]$$

$$= 0.25\,[0.5984 + 1 + 0.1411 + 0.6972]$$

$$y_1^{(1)} = 0.6091$$

Second modified value of y at $x_1=3$ is

$$y_1^{(2)} = y_0 + \frac{h}{2}\left[f(x_0,y_0) + f(x_1, y_1^{(1)})\right]$$

$$= 0 + \frac{0.5}{2}\left[\sin 2.5 + \cos 0 + \sin 3 + \cos(0.6091)\right]$$

$$= 0.25\,[0.5984 + 1 + 0.1411 + 0.8201]$$

$$= 0.6399$$

Now we require $y_2 = y(x_2) = y(3.5)$

We consider $x_1 = 3$, $y_1 = 0.6399$, $h = 0.5$

$$y_2^{(E)} = y_1 + h f(x_1, y_1)$$

$$= 0.6399 + 0.5 [\sin 3 + \cos(0.6399)]$$

$$= 0.6399 + 0.5 [0.1411 + 0.8021]$$

$$y_2^{(E)} = 1.1115$$

The First modified value at $x_2 = 3.5$ is

$$y_2^{(1)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(E)})]$$

$$= 0.6399 + \frac{0.5}{2} [\sin 3 + \cos(0.6399) + \sin(3.5) + \cos(1.1115)]$$

$$= 0.6399 + 0.25 [0.1411 + 0.8021 + (-0.3507) + 0.4433]$$

$$y_2^{(1)} = 0.8988$$

Second modified value at $x_2 = 3.5$ is

$$y_2^{(2)} = y_1 + \frac{h}{2} [f(x_1, y_1) + f(x_2, y_2^{(1)})]$$

$$= 0.6399 + \frac{0.5}{2} [\sin 3 + \cos(0.6399) + \sin 3.5 + \cos(0.898$$

$$= 0.6399 + 0.25 [0.1411 + 0.8021 - 0.3507 + 0.6225]$$

$$= 0.9436$$

$\therefore y(3.5) = 0.9436.$